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SUPERSEDING
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June 11, 1985

FEDERAL SPECIFICATION SHEET

SEWING MACHINE, INDUSTRIAL; FLATBED, SINGLE NEEDLE, 301 STITCH, PARACHUTE REPAIR

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring the sewing machines described herein shall consist of this document and the issue in effect of Specification OO-S-256.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for single needle, flatbed, 301 stitch, industrial sewing machines used in the fabrication and repair of aerial delivery equipment.

1.2 Classification. The sewing machines shall be of the following classes as specified (see 6.2).

Class 1 - Light duty
Class 2 - Medium duty
Class 3 - Light-heavy duty
Class 4 - Heavy duty

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

AMSC N/A FSC 3530

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DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

6-1-9360 Sewing Machine, Industrial, Flatbed, Single Needle, 301 Stitch, Light Duty: Parachute Repair and Fabrication Test Requirements

6-1-9361 Sewing Machine, Industrial, Flatbed, Single Needle, 301 Stitch, Medium Duty: Parachute Repair and Fabrication Test Requirements

6-1-9362 Sewing Machine, Industrial, Flatbed, Single Needle, 301 Stitch, Light Heavy Duty: Parachute Repair and Fabrication Test Requirements

6-1-9363 Sewing Machine, Industrial, Flatbed, Single Needle, 301 Stitch, Heavy Duty: Parachute Repair and Fabrication Test Requirements

6-1-8129 Sewing Machine, Industrial, Single Needle, Heavy Duty: Stand, Bag Repair Kit Assembly

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Class 1, machine. The class 1 machine shall be reversible with the capability of sewing fabrics that range from light weight nylon and cotton duck to light weight tape and webbing and shall utilize metric needle sizes 110, 120, 130, and 140. The machine shall operate at speeds not less than 4000 or greater than 5000 stitches per minute. The machine shall be automatically lubricated. The mounting shall be fully submerged. Each machine shall be supplied with bobbin winder and a knee lifter. The machine shall be equipped with a hand operated reverse lever which is spring loaded to normally return to the forward position.

3.1.1 Class 1, machine feed. The class 1 machine shall have a reversible type A feed. The presser foot shall be hinged. The feed dog shall have three

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rows of teeth. The teeth shall include a center row and one row on each side of the needle opening. The teeth shall be spaced a minimum of 16 teeth to the inch.

3.1.2 Class 1, stitches. The class 1 machine shall produce a minimum of 6 to a unlimited number of stitches per inch.

3.1.3 Class 1, height under the presser foot. The class 1 machine shall provide a minimum height of 9/16 of an inch under the presser foot.

3.1.4 Class 1, working room to the right of the needle. The class 1 machine shall be furnished with a working room distance to the right of the needle of 10 inches minimum to 16 inches maximum.

3.1.5 Class 1, table top. The class 1 machine shall be furnished with a type II table top. The outside dimensions of the table top shall be 20 by 40 by 2 inches thick with a tolerance of ± 1/4 of an inch for each dimension.

3.1.6 Class 1, spare parts. The class 1 machine shall be furnished with the following spare parts in the quantity indicated:

Spare parts for class 1 machine

Item identification	Quantity
Presser bar	1
Presser boot holding screw	2
Needle bar thread guide	1
Needle bar thread guide set screw	1
Throat plate	1
Throat plate screw	2
Feed dog screw	2
Tension assembly (complete)	1
Face plate screw	1
Presser bar lifting lever	1
Hook retainer set screw sewing	1
Hook retainer set screw washer sewing	1
Bed side plate spring	1
Bed side plate spring screw	1
Thread guide	1
Hinged presser foot	1
Bobbin case	1
Bobbin metal	10
Sewing hook assembly	1
Thread take up spring	1

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Needle, round point, chrome finish

Metric sizes:

110	100
120	100
130	100
140	100

3.2 Class 2, machine. The class 2 machine shall be reversible with the capability of sewing fabrics that range from light weight to heavy weight nylon and cotton duck and light weight to medium weight tapes and webbing and shall utilize metric needle sizes 110, 125, 140 and 180. The machine shall operate at speeds not less than 2500 or greater than 3500 stitches per minute. The machine shall be manually or automatically lubricated. The mounting shall be fully submerged. Each machine shall be supplied with a bobbin winder and a knee lifter. The machine shall be equipped with a hand operated reverse lever which is spring loaded to normally return to the forward position.

3.2.1 Class 2, machine feed. The class 2 machine shall be equipped with a reversible type H feed. The feed dog shall have a center row of teeth. The spacing shall be a minimum of 16 teeth per inch. The throat plate shall also have a fixed row of teeth on each side of the needle opening. The spacing shall be 25 teeth to the inch.

3.2.2 Class 2, stitches. The class 2 machine shall produce a minimum of 4 to a unlimited number of stitches per inch.

3.2.3 Class 2, height under the presser foot. The class 2 machine shall provide a minimum height of 0.55 of an inch under the presser foot.

3.2.4 Class 2, working room to the right of the needle. The class 2 machine shall be furnished with a working room distance to the right of the needle of 10 inches minimum to 18 inches maximum.

3.2.5 Class 2, table top. The class 2 machine shall be furnished with a type II table top. The outside dimensions shall be 20 by 40 by 2 inches thick with a $\pm$1/4 of an inch tolerance for each dimension.

3.2.6 Class 2, spare parts. The class 2 machine shall be furnished with the following spare parts in the quantity indicated:

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Spare parts for class 2 machine

Item Identification	Quantity
Needle bar thread guide	1
Needle bar thread guide screw	1
Thread tension thumb nut	1
Thread controller stud	1
Thread controller disc screw	1
Safety clutch latch spring	1
Bobbin tension spring	1
Bobbin case thread tension adjustment screw	2
Bobbin case tension spring molding screw	2
Gibb screw	3
Tension release slide	1
Feed dog screw	2
Throat plate screw	2
Bobbin, metal	10
Hook assembly sewing	1
Bobbin case	1
Thread take up spring	1
Needle round point, chrome finish	
Metric sizes:	
110	100
125	100
140	100
180	100

3.3 Class 3, machine. The class 3 machine shall be capable of sewing fabrics that range from medium to heavy nylon and cotton duck to multiple layers of medium to heavy webbing and shall utilize metric needle sizes 180, 230, and 250. The machine shall operate at speeds not less than 1000 or greater than 1500 stitches per minute. The machine shall be manually lubricated. The mounting shall be fully submerged. Each machine shall be supplied with a bobbin winder and foot lifter.

3.3.1 Class 3, machine feed. The class 3 machine shall have a type B feed. The feed dog shall have two double rows of teeth, one double row on each side of the needle opening. The rows shall have a minimum of 12 teeth per inch.

3.3.2 Class 3, stitches. The class 3 machine shall provide 2 to 13 stitches per inch.

3.3.3 Class 3, height under the presser foot. The class 3 machine shall provide a minimum height of 33/64 of an inch under the presser foot with needle and feed dogs in all positions.

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3.3.4 Class 3, working room to the right of the needle. The class 3 machine shall be furnished with a working room distance to the right of the needle of 10 inches minimum to 20 inches maximum.

3.3.5 Class 3, table top. The class 3 machine shall be furnished with a type II table top. The outside dimensions shall be 20 by 40 by 2 inches thick with a ±1/4 of an inch tolerance for each dimension.

3.3.6 Class 3, spare parts. The class 3 machine shall be furnished with the following spare parts in the quantity indicated:

Spare parts for class 3 machine

Item Identification	Quantity
Slide plate spring	1
Slide plate spring screw	1
Feed dog	1
Feed dog screw	2
Rotating hook with bobbin case	1
Bobbin case position bracket	1
Thread controller spring	1
Thread take up spring	1
Thread tension thumb nut	1
Face plate screw (upper)	1
Face plate screw (lower)	1
Outside presser foot	1
Outside presser foot screw	1
Inside presser foot screw	1
Inside presser foot	1
Needle bar pinch screw	1
Needle bar needle clamp	1
Needle bar thread guide (lower)	1
Needle bar thread guide screw	1
Throat plate	1
Throat plate screw	2
Needle, round point, chrome finish	
Metric sizes:	
180	100
230	100
250	100

3.4 Class 4, machine. The class 4 machine shall be reversible and capable of sewing fabrics that range from heavy cotton and nylon duck to multiple layers of heavy webbing and shall utilize metric needle sizes 230, 250, 280, 300, and 330. The machine shall operate at speeds not less than 500

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or greater than 800 stitches per minute. The machine shall be manually lubricated. The mounting shall be fully submerged. Each machine shall be supplied with a bobbin winder and a foot lifter. The machine shall be equipped with a hand operated reverse lever which is spring loaded to normally return to the forward position.

3.4.1 Class 4, machine feed. The class 4 machine shall have a type B feed. The feed dog shall have two double rows of teeth. One double row on each side of the needle opening. The rows shall have a minimum of 7 teeth per inch.

3.4.2 Class 4, stitches. The class 4 machine shall provide a minimum of 2 to an unlimited number of stitches per inch.

3.4.3 Class 4, height under the presser foot. The class 4 machine shall provide a minimum height of 1 inch under the presser foot.

3.4.4 Class 4, working room to the right of the needle. The class 4 machine shall be furnished with a working room distance to the right of the needle of 16-1/4 inches minimum to 20 inches maximum.

3.4.5 Class 4, bed dimensions. The class 4 machine bed dimensions shall measure 10 by 26-1/4 by 6 inches deep ±1/4 of an inch tolerance for each dimension.

3.4.6 Class 4, stand and table top. Each class 4 machine shall be furnished with a table top and stand conforming to Drawing 6-1-8129.

3.4.7 Class 4, spare parts. The class 4 machine shall be furnished with the following spare parts in the quantity indicated:

Spare parts for class 4 machine

Item Identification	Quantity
Feed dog	1
Throat plate screw	2
Needle plate strip screw	2
Cloth plate screw	4
Shuttle race holding set screw	2
Shuttle race frame	1
Needle clamp	1
Needle clamp holding screw	1
Thread tension thumb nut	1
Thread tension spring	1
Thread controller bushing set screw	1
Bobbin case opener tool	1
Rock shaft nut	1
Vibrating presser bar link	1
Needle plate strip	1

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Spare parts for class 4 machine (cont'd)

Item Identification	Quantity
Shuttle race	1
Lifting presser foot	1
Vibrating presser foot	1
Thread take up check spring	1
Shuttle body complete	1
Bobbin	10
Needle round, point, chrome finish	
Metric sizes:	
230	100
250	100
280	100
300	100
330	100

3.5 Performance.

3.5.1 Class 1 machine, operation. The class 1 machine shall fabricate and install parachute seams, skirts, vee tabs, vent bands, and vent band splices when tested as specified in 4.1.1 and 4.2.1. The machine shall show no sign of stalling, pounding, shearing of threads, failure to lock stitches, or deformation of any part during the fabrication of these articles.

3.5.2 Class 2 machine, operation. The class 2 machine shall fabricate and install personnel and cargo parachute deployment bags, bridal loops, end slot reinforcements, basic patches, tie loops and tie loop reinforcement when tested as specified in 4.1.2 and 4.2.2. The machine shall show no signs of stalling, pounding, shearing of threads, failure to lock stitches, or deformation of any part during the fabrication of these articles.

3.5.3 Class 3 machine, operation. The class 3 machine shall fabricate and install personnel, cargo and extraction parachute, bridal loops, static lines, upper lateral bands and upper lateral band splices when tested as specified in 4.1.3 and 4.2.3. The machine shall show no signs of stalling, pounding, shearing of threads, failure to lock stitches, or deformation of any part during the fabrication of these articles.

3.5.4 Class 4 machine, operation. The class 4 machine shall fabricate and install personnel, cargo and extraction parachute, bridal loops, locking stow loops, harnesses and multiple layer heavy webbing splices when tested as specified in 4.1.4 and 4.2.4. The machine shall show no signs of stalling, pounding, shearing of threads, failure to lock stitches, or deformation of any part during the fabrication of these articles.

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4. QUALITY ASSURANCE PROVISIONS

4.1 First article testing. When a first article is required it shall be tested in accordance with the applicable test method specified in table I.

TABLE I. First article tests

Sewing Machine, class	Requirement paragraph	Test method
1	3.5.1	4.1.1
2	3.5.2	4.1.2
3	3.5.3	4.1.3
4	3.5.4	4.1.4

4.1.1 Class 1 machine test. The class 1 machine shall be operated to fabricate and assemble all of the typical parachute details as shown on Drawing 6-1-9360. Any nonconformance with the requirements of 3.5.1 shall constitute failure of this test.

4.1.2 Class 2 machine test. The class 2 machine shall be operated to fabricate and assemble all of the typical parachute details as shown on Drawing 6-1-9361. Any nonconformance with the requirements of 3.5.2 shall constitute failure of this test.

4.1.3 Class 3 machine test. The class 3 machine shall be operated to fabricate and assemble all of the typical parachute details as shown on Drawing 6-1-9362. Any nonconformance with the requirements of 3.5.3 shall constitute failure of this test.

4.1.4 Class 4 machine test. The class 4 machine shall be operated to fabricate and assemble all of the typical parachute details as shown on Drawing 6-1-9363. Any nonconformance with the requirements of 3.5.4 shall constitute failure of this test.

4.2 End item testing. Each end item shall be tested in accordance with the applicable test method specified in table II. Failure of any test shall be cause for rejection of the end item involved.

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TABLE II. End item testing

Sewing Machine, class	Requirement paragraph	Test method
1	3.5.1	4.2.1
2	3.5.2	4.2.2
3	3.5.3	4.2.3
4	3.5.4	4.2.4

4.2.1 Class 1 machine test. Each class 1 machine shall be operated to sew samples of typical parachute construction, as shown by details F and G on Drawing 6-1-9360. Any nonconformance with the requirements of 3.5.1 shall constitute failure of this test. The complete test samples shall be supplied with each machine furnished.

4.2.2 Class 2 machine test. Each class 2 machine shall be operated to sew samples of typical parachute construction, as shown by details F and G on Drawing 6-1-9361. Any nonconformance with the requirements of 3.5.2 shall constitute failure of this test. The complete test samples shall be supplied with each machine furnished.

4.2.3 Class 3 machine test. Each class 3 machine shall be operated to sew samples of typical parachute construction, as shown by details F and G on Drawing 6-1-9362. Any nonconformance with the requirements of 3.5.3 shall constitute failure of this test. The complete test samples shall be supplied with each machine furnished.

4.2.4 Class 4 machine test. Each class 4 machine shall be operated to sew samples of typical parachute construction, as shown by details F and G on Drawing 6-1-9363. Any nonconformance with the requirements of 3.5.4 shall constitute failure of this test. The complete test samples shall be supplied with each machine furnished.

5. PACKAGING

Packaging shall be as specified in the basic specification and in the contract or purchase order.

6. NOTES

6.1 Intended use. The sewing machines are intended for industrial use for fabricating and repairing parachutes.

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6.2 Ordering data.

a. Title, number, and date of this specification.
b. Class of machine required (see 1.2).

MILITARY INTERESTS:

Custodians:

Army - GL
Air Force - 99

Review activity:

DLA - GS

User activity:

Air Force - 50

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

Project No. 3530-0078

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